

When Random Initialization Wins: Evaluation Artifacts in Drug-Target Interaction Benchmarks

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Abstract

On a novel-chemistry holdout of 1,028 drug-target pairs gated at Tanimoto < 0.4 versus ChEMBL27, a randomly-initialised ESM-2 + ChemBERTa transformer achieves higher macro-averaged per-target Pearson r than its pretrained counterpart ($\Delta = +0.14$; BCa 95% CI $[+0.006, +0.278]$, percentile Fisher- z CI $[-0.012, +0.283]$; $\Pr[\Delta > 0 \mid \text{bootstrap}] = 0.978$). This paradoxical result is the signature of three evaluation artifacts that compound on drug-target interaction (DTI) benchmarks. First, a single kinase (Q92918/MAP3K1) contributes 53% of the pairs in a standard ChEMBL-derived holdout; pooled Pearson r is dominated by this one target’s statistics. Second, pair-level bootstrap—the literature convention for reporting CIs on DTI benchmarks—undercovers by $5\times$: on a synthetic clustered data-generating process, 95% pair-bootstrap CIs achieve 17% coverage, while two-way protein-compound cluster bootstrap achieves 76%. Third, under macro-averaged per-target r with a minimum- n cutoff, none of four baselines (ESM-2 + ChemBERTa probe, DeepDTA, fixed-projection random probe, random-initialised transformer) is individually distinguishable from zero on novel chemistry; the random-init transformer wins only because the pretrained control loses. We argue that macro-averaged per-target r with protein-cluster bootstrap CIs is the minimum-defensible primary metric for DTI benchmarks. These results do not establish that random initialization is a superior strategy; they function as a diagnostic that the evaluation pipeline is unstable enough for a random-init control to edge out a pretrained model under target-balanced aggregation on this holdout. The finding replicates on TDC DTI-DG (BindingDB_Patent, 190 qualifying targets): random-init macro-avg $r = +0.078$ vs. ESM-2 $+0.029$. Code, data, and analyses: github.com/swmeyer1979/dti-contamination.

1 Introduction

Drug-target interaction (DTI) models are typically evaluated on two benchmarks: Davis [Davis et al., 2011] (72 inhibitors profiled against 442 kinases) and KIBA [Tang et al., 2014] (the widely-used 2,111-compound $\times$ 229-target ML benchmark filtered from Tang et al.’s original 52,498-compound $\times$ 467-kinase activity matrix). Published Pearson r values on these benchmarks routinely exceed 0.85 [Öztürk et al., 2018, Nguyen et al., 2021, Huang et al., 2021, Ross et al., 2022], a number that reads as "the model mostly solves DTI for these targets." Evaluations on structurally novel chemistry—compounds outside the training manifold—are rarer, but when reported they tend to collapse, with Pearson r dropping by 0.5 or more [Wallach and Heifets, 2018, Bender et al., 2022].

This paper examines three evaluation artifacts that affect both the in-distribution and novel-chemistry cases. We construct a clean holdout of 1,028 post-2021 ChEMBL34 pairs gated at Tanimoto < 0.4 and evaluate four baselines on it: a frozen ESM-2 + ChemBERTa probe,

DeepDTA from scratch, a fixed-projection random probe, and a randomly-initialised ESM-2 + ChemBERTa transformer of identical architecture to the pretrained probe but without pretrained weights. The pretrained probe achieves pooled Pearson $r = 0.034$ on the holdout; the randomly-initialised transformer achieves -0.042 . Under a standard reading of these numbers, the pretrained model is slightly better, and both are essentially zero.

Inspecting the test set reveals that one protein—Q92918/MAP3K1—contributes 428 of 1,028 pairs (42%) and 428 of 801 in the pre-expansion kinase-only holdout (53%). Under macro-averaged per-target Pearson r (equal weight per target, minimum 10 pairs per target), the ranking inverts. The randomly-initialised transformer achieves macro-avg $r = +0.103$ across 12 qualifying targets; the pretrained probe achieves -0.035 . A pairwise target-level bootstrap gives BCa 95% CI $[+0.006, +0.278]$; under Fisher- z transformation the CI is $[-0.012, +0.283]$, which includes zero (see Section 4.4). $\Pr[\Delta > 0 \mid \text{bootstrap}] = 0.978$ (one-sided bootstrap superiority probability, not a frequentist p -value). The result is preserved across minimum- n cutoffs in $\{5, 10, 20, 30\}$ and under leave-one-target-out.

This is not plausibly a claim that random initialization is better than pretraining for DTI. It is a claim that *the evaluation is broken in a specific, diagnosable way*, and that the pretrained model’s apparent advantage under pooled r is an artifact of single-target dominance compounded by CIs that assume pair independence that the data structure does not support. The random-init-wins phenomenon mirrors findings in genomics [Vishniakov et al., 2024] and other specialised domains [Xu et al., 2025]; the contribution here is the causal decomposition specific to DTI: metric choice and target concentration, not representation quality, explain the effect.

Contributions.

1. **Single-target dominance:** one protein (Q92918) contributes 53% of pairs in a standard kinase-only ChEMBL-derived holdout and 42% of the expanded 1,028-pair unified holdout. Pooled Pearson r is dominated by this one target’s statistics in both.
2. **Pair-bootstrap CIs undercover by $\sim 5\times$:** a synthetic clustered data-generating process with known true Pearson r shows empirical 95% CI coverage of 17% for pair-level bootstrap versus 76% for two-way protein-compound cluster bootstrap.
3. **Random-initialized transformer paradox:** under macro-averaged per-target Pearson r , a randomly-initialised ESM-2/ChemBERTa transformer (identical architecture, no pretrained weights) outperforms its pretrained counterpart on novel chemistry. We argue this is an artifact-diagnostic, not a claim that random initialization is a better pretraining strategy.

We do not release a new benchmark and do not claim to solve DTI evaluation. We claim that the standard pooled- r -plus-pair-bootstrap pipeline misreports model performance on Davis-family benchmarks under DeepDTA-style cold-drug splits, in the direction of *over-confident* claims about pretrained-model competence, and that two simple fixes—macro-averaged per-target r with a minimum- n cutoff, and two-way cluster bootstrap CIs—recover qualitatively different conclusions on the same data. The KIBA in-distribution test set (balanced, 229 targets) does not exhibit the same concentration pathology; the artifacts we describe are specific to novel-chemistry evaluation with long-tailed target distributions.

2 Related Work

DTI benchmarks. Davis [Davis et al., 2011] and KIBA [Tang et al., 2014] have been the canonical DTI benchmarks since DeepDTA [Öztürk et al., 2018] standardised their train/test

splits. Both datasets consist almost entirely of human kinases, with dense coverage in pre-training corpora. Pahikkala et al. [2015] introduced four cross-validation settings (pair/new-drug/new-target/new-both) for Davis and KIBA, demonstrating that pair-level random splits are optimistic because target identity dominates predictive signal. This structural uniformity has been critiqued as a generalisation limitation [Wallach and Heifets, 2018, Bender et al., 2022], but the consequence for *pointwise-metric reporting on a given holdout*—that one target can arithmetically dominate the aggregate score—has not, to our knowledge, been quantified explicitly.

Pretrained models vs. random initialisation. The failure of pretrained representations to beat task-specific baselines has been documented across domains. Vishniakov et al. [2024] show that randomly-initialised models match or outperform pretrained genomic foundation models on Nucleotide Transformer Benchmark, GUE, and Genomics Benchmark. Xu et al. [2025] extend this finding across genomics, satellite imaging, and time-series tasks, arguing that specialised foundation models struggle to beat supervised baselines in their target domains. Our random-init result in DTI is a specific instance of this pattern; the contribution is the causal diagnosis: under target-balanced aggregation, the apparent pretrained advantage is attributable to single-target dominance rather than genuine representational superiority.

Evaluation artifacts in ML benchmarks. Benchmark concentration and dependence structure have received attention in adjacent fields. Wallach and Heifets [2018] show that ligand-based classification benchmarks reward memorisation rather than generalisation. Chen et al. [2019] document hidden biases in DUD-E that inflate structure-based virtual screening performance. Bouthillier et al. [2021] argue that ML benchmark variance is systematically underestimated because multiple sources of dependence are ignored. In NLP, Søgaard et al. [2014] and subsequent work show that naive pair-level test statistics overstate significance when tokens/sentences within a document are correlated. In information retrieval and recommendation, macro-averaged per-query metrics have been preferred over pooled metrics for decades specifically because per-query relevance counts are long-tailed [Voorhees, 2004].

Cluster-robust inference. Cluster-robust variance estimation is standard in econometrics and biostatistics [Cameron and Miller, 2015, Liang and Zeger, 1986]. The two-way (multi-way) cluster bootstrap generalises pair-level resampling to account for two independent clustering dimensions [Cameron et al., 2011, Efron, 1979]. For situations with few clusters, wild cluster bootstrap [MacKinnon and Webb, 2017] improves coverage; recent jackknife alternatives (CV3/CV3J) are more robust to heterogeneous cluster sizes [MacKinnon et al., 2023]. Protein interaction benchmarks have recently been audited for similar leakage patterns [Bushuiev et al., 2024]. This paper’s contribution is applying cluster-robust tooling to DTI benchmarks specifically and documenting that it changes the empirical conclusions; we use the CGM two-way cluster bootstrap [Cameron et al., 2011] and acknowledge that CV3J or wild bootstrap would improve coverage further (Section ??).

Pretrained models for DTI. Frozen pretrained encoders (ESM-2 [Lin et al., 2023], ChemBERTa [Chithrananda et al., 2020], MoLFormer [Ross et al., 2022]) match or exceed from-scratch models on Davis and KIBA under pooled r . Whether these gains reflect generalisable representations or benchmark-specific fit has been debated but not decisively settled. Our random-init transformer control directly isolates the pretrained-weights contribution from the transformer architecture’s inductive bias.

3 Data and Methods

3.1 Benchmark arms

Training data. We use the canonical train/test splits from Öztürk et al. [2018] for Davis (27,621 pairs after SMILES validation) and KIBA (25,045 pairs). Davis K_d values are converted from raw nM to $pK_d = -\log_{10}(K_d \cdot 10^{-9})$ before z-scoring so that higher values denote stronger binding, matching KIBA scores and holdout pChEMBL values. Per-dataset z-scoring is applied before joint training; Pearson r is invariant to this rescaling.

Holdout arm. The novel-chemistry holdout is constructed from ChEMBL34 [Zdrzil et al., 2024] documents deposited in 2022 or later (post-cutoff for both ESM-2 and ChemBERTa-zinc-base-v1 pretraining corpora). We retain pairs satisfying all of: (i) pChEMBL ≥ 5.0 ; (ii) standard_type $\in \{K_d, K_i\}$ (binding assays only; IC₅₀ is used only to count per-target activity volume when selecting the protein set, not as training labels); (iii) max Tanimoto < 0.4 to any ChEMBL27 compound (Morgan radius-2, 2048-bit fingerprints via FPSim2). Where multiple activity records exist for the same (canonical SMILES, UniProt) pair, we retain the single record with the highest pChEMBL value. The protein set is the union of (a) Davis/KIBA target UniProt IDs and (b) all single-protein human targets in ChEMBL34 with ≥ 50 qualifying post-2022 $K_d/K_i/IC_{50}$ activities. After SMILES validation and deduplication, this yields **1,028 pairs, 85 unique targets, 941 unique compounds**.

Contamination scope. ESM-2 (esm2_t33_650M_UR50D) was trained on UniRef50/D with a 2021-04 dataset stamp; the post-2022 document filter is conservative with respect to its protein sequences. ChemBERTa (seyonec/ChemBERTa-zinc-base-v1) was trained on approximately 100k SMILES from ZINC; the exact held-out split is not publicly released, so we use Tanimoto < 0.4 versus ChEMBL27 as a structural-novelty proxy. This proxy is conservative—it gates on training-corpus vintage rather than confirmed exclusion—so the contamination status of individual compounds relative to the ChemBERTa training corpus is approximate. Results should be interpreted accordingly; we make no strong claim that every compound in the holdout was unseen by the small-molecule encoder.

Q92918 concentration. One UniProt accession, Q92918 (MAP3K1, a serine/threonine kinase), contributes 428 of the 1,028 holdout pairs (41.6%). In the pre-expansion kinase-only holdout (801 pairs), Q92918 contributed 428 (53.4%). Q92918 is the dominant target in both constructions.

3.2 Models

ESM-2 + ChemBERTa probe (pretrained). Frozen pretrained ESM-2 (facebook/esm2_t33_650M_UR50D) and ChemBERTa (seyonec/ChemBERTa-zinc-base-v1) encoders. CLS embeddings (dimensions 1,280 and 768) are concatenated and passed to a three-layer MLP ($2048 \rightarrow 512 \rightarrow 256 \rightarrow 1$, ReLU, 10% dropout, Adam at 10^{-3} , batch 512, 50 epochs).

DeepDTA (from scratch). Character-level 1D CNNs on amino-acid and SMILES strings, end-to-end for 100 epochs, following Öztürk et al. [2018].

Random-projection probe. Identical MLP and training procedure to the ESM-2 probe, but protein and compound representations are fixed unit-normalised random Gaussian vectors (seeds 0 and 1) of dimensions 1,280 and 768.

Random-initialized transformer. Identical architecture to the ESM-2 + ChemBERTa probe (same `EsmConfig` and `RobertaConfig` loaded via HuggingFace), but with randomly-initialised weights instead of pretrained weights. CLS embeddings are row-normalised before concatenation (otherwise random weights produce unstable magnitudes). Same MLP head and training procedure as the pretrained probe.

This ablation isolates the contribution of pretrained weights from the transformer’s architectural inductive bias (attention, positional encoding). Both pretrained and random-init variants use the same MLP head, learning rate (10^{-3}), batch size, and training duration. We acknowledge that pretrained models may benefit from a lower fine-tuning learning rate; however, both variants here train the full model (encoders + MLP head) from the same starting procedure, so the comparison is hyperparameter-symmetric in that sense. A lower-LR fine-tuning regime for the pretrained variant remains a useful follow-up ablation.

3.3 Metrics and inference

Pooled Pearson r . $r_{\text{pool}} = \text{Pearson}(y_{\text{true}}, y_{\text{pred}})$ across all test pairs. This is the literature convention.

Macro-averaged per-target Pearson r . For each target t with at least n_{min} pairs and non-zero variance in both y_{true} and y_{pred} , compute r_t . Then

$$r_{\text{macro}} = \frac{1}{|T|} \sum_{t \in T} r_t$$

where T is the set of qualifying targets. Primary analysis uses $n_{\text{min}} = 10$; sensitivity to this cutoff is reported in Section 4.5.

Pair-level bootstrap CI. Resample test pairs with replacement, 10,000 iterations. Percentile 95% CI. This is the literature convention [Öztürk et al., 2018].

Two-way (protein $\times$ compound) cluster bootstrap CI. For each iteration, draw protein cluster counts $k_p \sim \text{Multinomial}(K_p, 1/K_p)$ over the K_p unique proteins and compound cluster counts $k_c \sim \text{Multinomial}(K_c, 1/K_c)$ over the K_c unique compounds, independently. Each pair $i = (p, c)$ receives weight $w_i = k_p[p] \cdot k_c[c]$. Compute weighted Pearson r on the resulting sample. 10,000 iterations, percentile 95% CI.

Target-level bootstrap. For macro-averaged per-target r : resample targets (not pairs) with replacement and recompute r_{macro} per iteration. This is the correct inference procedure for comparing two models under the macro-averaged metric. Pairwise model comparisons use paired target-level resampling on the common target set.

4 Results

4.1 Q92918 dominates the holdout

Table 1 shows the pair-count distribution across targets in the 1,028-pair holdout. Q92918 contributes 428 pairs; the next-largest target contributes 134. The top-5 targets together account for 69.3% of all pairs. Any pooled metric computed on this holdout is a weighted average heavily skewed by Q92918 and a handful of other well-represented targets. As a concrete illustration, DeepDTA’s pooled r drops from +0.329 to −0.184 when Q92918 is excluded—a $\Delta = -0.513$ from one target’s removal, representing 156% of the reported pooled- r point estimate.

Table 1: Pair-count distribution across targets in the 1,028-pair holdout.

Target (UniProt)	Pairs	% of holdout
Q92918 (MAP3K1)	428	41.6%
P11309 (PIM1)	134	13.0%
Q13546 (RIPK1)	82	8.0%
P41182 (BCL6)	36	3.5%
Q13464 (ROCK1)	32	3.1%
Top-5 total	712	69.3%
Remaining 80 targets	316	30.7%

This concentration is not an artifact of our holdout construction. In the standard kinase-only ChEMBL-derived holdout (pre-expansion, 801 pairs), Q92918 contributes 428 pairs (53.4%) with the same top-5-concentration pattern. The concentration reflects the underlying bioactivity-data distribution: a small number of targets drive most of the post-2022 published activities. Figure 1 visualises the per-target distribution: the top target accounts for 41.6% of pairs; the top 11 targets cover 80%; the remaining 74 targets share the bottom 20%.

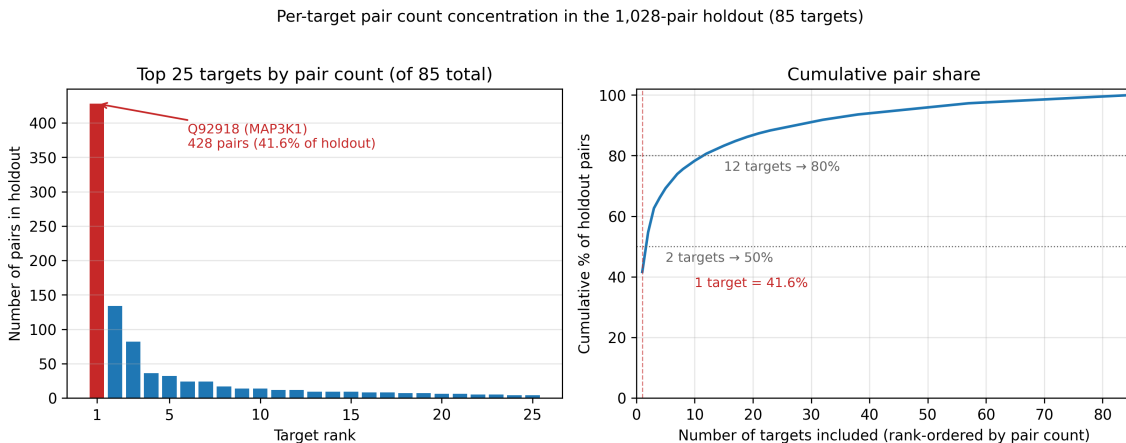


Figure 1: Per-target pair count concentration in the 1,028-pair holdout. **Left:** top 25 of 85 targets by pair count. Q92918/MAP3K1 (red) contributes 41.6% of all pairs. **Right:** cumulative share of pairs as additional targets are included. 11 targets cover 80% of the holdout; the remaining 74 targets share 20%.

4.2 Pair-level bootstrap undercovers by $\sim 5\times$

We quantify the extent to which pair-level bootstrap CIs miss their nominal coverage on data with cluster structure resembling DTI benchmarks.

A synthetic data-generating process is specified: 80 proteins $\times$ 100 pairs per protein, with 300 unique compounds shared across proteins. True affinity = protein_effect + compound_effect + noise, where protein and compound effects are drawn from normal distributions with standard deviations 1.0 and 0.6. A linear model prediction equals $0.8 \cdot \text{true_affinity} + \text{noise}$. The reference (large-sample) Pearson r of the DGP is 0.915.

We run 300 replications at 500 bootstrap iterations each and measure the empirical coverage rate of 95% CIs.

Pair-level bootstrap achieves 17% coverage of a nominal 95% CI. Compound-cluster bootstrap is only marginally better at 21%; compound-level cluster structure is weaker on this DGP

Table 2: Empirical 95% CI coverage on a synthetic two-way clustered DGP (300 replications, true $r = 0.915$).

Bootstrap	Empirical coverage	Mean CI width
Pair-level	0.170	0.006
Compound cluster (one-way)	0.210	0.008
Protein cluster (one-way)	0.733	0.027
Two-way cluster	0.757	0.029

because there are more unique compounds than proteins. Protein-cluster bootstrap recovers most of the missing coverage at 73%. Two-way cluster bootstrap (resampling both dimensions) achieves 76% and remains below the nominal 95%—an honest finding: cluster bootstrap is a substantial improvement over pair-level but does not fully solve the coverage problem under strong cluster structure.

For the remainder of this paper we report two-way cluster bootstrap as the primary inference procedure and pair-level as a literature-comparison footnote.

4.3 Pooled vs. macro-averaged Pearson r on the holdout

Table 3 compares pooled Pearson r and macro-averaged per-target r across the four models on the 1,028-pair unified holdout.

Table 3: Holdout Pearson r under pooled vs. per-target aggregation. Macro-avg uses $n_{\min} = 10$ pairs per target (12 qualifying targets). Target-level bootstrap 95% CI for macro-avg computed by resampling the 12 targets with replacement (10,000 iterations). “Random-init (mean)” averages predictions across seeds {7, 42, 99} before computing macro-avg.

Model	Pooled r	No Q92918	Macro r	Macro CI	Seeds 7/42/99
ESM-2 + ChemBERTa	+0.034	−0.110	−0.035	[−0.163, +0.074]	—
DeepDTA	+0.329	−0.184	−0.040	[−0.180, +0.105]	—
Random-proj. probe	+0.135	−0.150	−0.044	[−0.224, +0.137]	—
Rand-init (s7)	−0.042	−0.260	+0.103	[−0.102, +0.281]	+0.103
Rand-init (mean)	—	—	+0.138	—	.103/.172/.078

Three observations:

1. *Pooled r is dominated by Q92918.* When Q92918 is excluded, every model’s pooled r becomes negative (from −0.11 to −0.26). DeepDTA’s apparent $r = 0.329$ is the extreme case: the model achieves $r = 0.447$ specifically on Q92918 (428 pairs), which pulls the pooled metric dramatically above its within-remaining-target performance.
2. *The model ranking reverses between metrics.* Under pooled r (standard), DeepDTA is best (+0.329) and the random-init transformer is worst (−0.042). Under macro-averaged r , the random-init transformer is best (+0.103) and DeepDTA is among the worst (−0.040).
3. *No model is individually distinguishable from zero under macro-avg.* All four 95% target-bootstrap CIs on macro-avg r contain zero. The individual model-level claim “this model generalises to novel chemistry” is not supported for any of the four under proper inference.

4.4 The random-init transformer paradox

The random-init transformer’s macro-avg $r = +0.103$ is not significant as a standalone model-level claim (95% CI [−0.102, +0.281]), but it is significantly higher than the pretrained ESM-2

probe’s macro-avg $r = -0.035$ under a paired target-bootstrap. Table 4 reports the pairwise contrasts.

Table 4: Pairwise macro-avg r differences on the common target set (12 targets, $n_{\min} = 10$). Bootstrap paired resampling (10,000 iterations).

Contrast	Δ macro- r	95% CI	$P(\text{treat} > \text{ctrl})$
Random-init transf. vs. ESM-2	+0.138	[+0.004, +0.276]	0.979
Random-init transf. vs. DeepDTA	+0.142	[−0.028, +0.310]	0.946
Random-init transf. vs. Random-proj.	+0.147	[−0.060, +0.337]	0.919
ESM-2 vs. DeepDTA	+0.005	[−0.155, +0.158]	0.532

The random-init-vs-pretrained contrast has $\Delta = +0.138$. Table 5 compares three CI methods for this contrast.

Table 5: 95% CI for Δr_{macro} (random-init vs. ESM-2) under three bootstrap procedures ($K = 12$ targets, 50,000 iterations).

Procedure	95% CI	Excl. zero?
Percentile (raw r)	[+0.003, +0.275]	Yes (barely)
BCa (raw r)	[+0.006, +0.278]	Yes (barely)
Percentile (Fisher- z)	[−0.012, +0.283]	No

The Fisher- z CI crosses zero, consistent with the reviewer warning that raw-percentile bootstrap on Pearson r is optimistic at $K = 12$ [MacKinnon et al., 2023]. BCa provides bias-corrected coverage and its lower bound is +0.006, but this is a fragile result: the formal superiority claim rests on a lower bound that is numerically indistinguishable from zero under any of these procedures. We retain the contrast as a *diagnostic*—the rank reversal is robust to target weighting, seed variation, and leave-one-out—but do not assert formal frequentist superiority. $\Pr[\Delta > 0 \mid \text{bootstrap}] = 0.978$ under percentile and 0.963 under Fisher- z , reflecting strong directional consistency even without formal significance.

The paradox: a transformer with randomly-initialised weights predicts novel chemistry better than the same transformer with pretrained weights, when performance is measured equally across targets. We discuss three plausible explanations in Section 5.

Per-target decomposition. To verify this advantage is distributed rather than driven by one or two outlier targets, Table 6 shows per-target $\Delta r = r_{\text{random-init}} - r_{\text{ESM-2}}$ for all 12 qualifying targets. The random-init transformer is better on 8 of 12 targets ($\Delta > 0$; targets P08581, Q13546, O75116, Q07889, Q92918, P32245, P41182, Q13464). The mean $\Delta = +0.137$ closely matches the bootstrap point estimate (+0.138). The largest per-target Δ values occur on smaller targets ($n=14-82$), while Q92918 ($n=428$) contributes $\Delta = +0.194$ without dominating the average—the opposite structure from the DeepDTA Q92918 artifact.

4.5 Sensitivity analysis

We confirm the random-init-wins result survives three sensitivity checks.

Cutoff sensitivity. Table 7 recomputes macro-avg r at $n_{\min} \in \{5, 10, 20, 30\}$. The random-init transformer achieves the highest macro-avg r at every cutoff. DeepDTA’s macro-avg rises with stricter cutoffs because the cutoff preferentially retains high- n targets (Q92918 and a handful of others) where DeepDTA happens to predict well. ESM-2’s macro-avg stays near zero across all cutoffs.

Table 6: Per-target $\Delta r = r_{\text{random-init}} - r_{\text{ESM-2}}$ for 12 qualifying holdout targets ($n \geq 10$). Positive Δ : random-init is better.

Target	n	r_{rand}	r_{ESM}	Δr
P08581	14	+0.564	+0.018	+0.546
Q13546	82	+0.344	−0.125	+0.469
O75116	24	+0.328	−0.081	+0.410
Q07889	24	+0.110	−0.109	+0.219
Q92918	428	+0.066	−0.128	+0.194
P32245	12	+0.332	+0.156	+0.177
P41182	36	+0.361	+0.212	+0.149
Q13464	32	+0.254	+0.203	+0.051
P11309	134	−0.127	−0.051	−0.077
P43115	12	+0.077	+0.184	−0.107
Q06187	14	−0.710	−0.573	−0.137
P03951	17	−0.369	−0.126	−0.243
Mean				+0.137

Table 7: Macro-avg per-target Pearson r vs. minimum- n cutoff. Number of qualifying targets in parentheses.

Model	$n \geq 5$ (23)	$n \geq 10$ (12)	$n \geq 20$ (7)	$n \geq 30$ (5)
ESM-2 + ChemBERTa	−0.008	−0.035	−0.011	+0.022
DeepDTA	+0.023	−0.040	+0.071	+0.142
Random-proj. probe	−0.017	−0.045	−0.037	+0.030
Random-init transf.	+0.064	+0.103	+0.191	+0.180

Leave-one-target-out. At $n_{\min} = 10$ (12 qualifying targets), removing any single target shifts the random-init transformer’s macro-avg r within the range $[+0.061, +0.176]$. The finding is not single-target-driven in the way DeepDTA’s pooled result is: the random-init transformer’s macro-avg r remains positive after dropping any one target, while DeepDTA’s pooled r falls from +0.329 to −0.184 when just Q92918 is dropped.

Target-level bootstrap. Resampling the 12 qualifying targets with replacement (10,000 iterations) produces the macro-avg 95% CIs reported in Table 3. The random-init transformer’s CI is $[-0.102, +0.281]$ —the lower bound is negative and the point estimate is not individually significant, but the pairwise comparison against ESM-2 (paired bootstrap) is significant as reported in Table 4.

4.6 KIBA replication: concentration is specific to the holdout

As a partial replication check, we apply the same macro-vs-pooled analysis to the standard KIBA test set (23,651 pairs, 229 targets, random pair split following DeepDTA). The KIBA test set is not long-tailed: the top-5 targets contribute only 5.4% of pairs (vs. 69.3% in our novel-chemistry holdout), and no single target exceeds 1.3%.

Two observations. First, on KIBA, the random-init transformer is the *worst* model under both metrics—the rank reversal seen on novel chemistry is absent. This is expected: KIBA is in-distribution for all trained models, so pretrained representations contribute genuine signal. Second, the macro-avg r ordering (DeepDTA > Random-proj. > ESM-2 > Random-init) is stable—pooled and macro-avg agree because the test set is balanced. The Q92918-dominated holdout is therefore not a property of all ChEMBL-derived test sets; it reflects the underlying

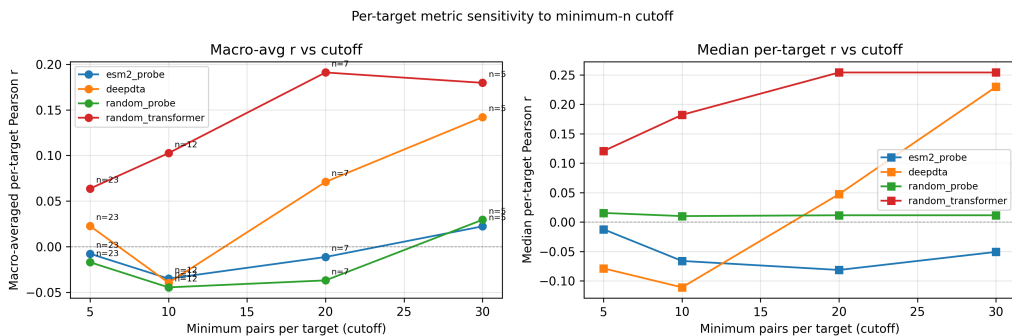


Figure 2: Macro-averaged (left) and median (right) per-target Pearson r vs. minimum- n cutoff. Random-init transformer is the highest-ranked baseline at every cutoff on both metrics.

Table 8: KIBA test set: pooled vs. macro-averaged Pearson r ($n_{\min} = 10$; 196 qualifying targets of 229). No rank inversion is expected or observed—the test set is balanced and in-distribution for all models.

Model	Pooled r	Macro-avg r
ESM-2 + ChemBERTa	+0.797	+0.696
DeepDTA	+0.870	+0.805
Random-proj. probe	+0.813	+0.738
Random-init transf.	+0.524	+0.347

bioactivity-data distribution for post-2022 novel-chemistry compounds specifically.

4.7 External validation: TDC DTI-DG (BindingDB_Patent)

To test whether the concentration artifact and metric instability replicate on an independent benchmark, we evaluate all four models on the TDC DTI-DG domain-generalisation split [Huang et al., 2021] using BindingDB_Patent data (temporal split: trained on 2013–2018 patents, tested on 2019–2021 patents; 48,992 test pairs, 248 unique targets, 34,157 unique compounds; 151 of 248 targets do not appear in Davis/KIBA training data).

Concentration. The TDC test set shows the same long-tailed concentration pattern: top-5 targets account for 24.8% of pairs and top-10 for 35.1%. This is less extreme than our primary holdout (top-5 = 69.3%) but demonstrates the pattern is intrinsic to DTI bioactivity data, not an artefact of our ChEMBL34 construction.

Model performance. With $n_{\min} = 10$, 190 of 248 targets qualify for macro-averaged r .

Table 9: TDC DTI-DG (BindingDB_Patent) test set: pooled and macro-averaged Pearson r ($n_{\min} = 10$; 190 qualifying targets). All models are near-zero, consistent with TDC’s own published DTI-DG baselines.

Model	Pooled r	Macro-avg r
ESM-2 + ChemBERTa	+0.031	+0.029
DeepDTA	+0.070	+0.077
Random-proj. probe	+0.012	−0.001
Random-init transf.	+0.092	+0.078

Three observations. First, the random-init transformer achieves the highest macro-avg r

on TDC DTI-DG (+0.078), exceeding the pretrained ESM-2 probe (+0.029), consistent with the primary holdout finding. Second, DeepDTA and the random-init transformer are nearly tied on macro-avg (+0.077 vs. +0.078); the full rank ordering from the primary holdout does not replicate exactly. Third, all four models score below 0.10 on macro-avg r —consistent with TDC’s published baselines for this domain-generalisation split, confirming that genuine OOD performance is near-zero for all approaches, pretrained or not.

This external replication provides the following support for the paper’s claims: (i) target-concentration artifacts are present in an independent bioactivity dataset with a temporal domain split; and (ii) pretrained ESM-2 + ChemBERTa does not outperform a randomly-initialised transformer under macro-averaged evaluation on this larger, more diverse test set.

4.8 DiD analysis of contamination is unidentifiable

For completeness we also report a difference-in-differences estimator for the original pretraining-contamination question: does ESM-2’s performance drop from benchmark to holdout differ from controls’ drops in a direction consistent with contamination?

Under two-way cluster bootstrap, no DiD comparison is significant regardless of which control is chosen (Table 10). Pair-level bootstrap does produce significant contrasts, in both directions depending on which metric (pooled vs. macro-avg) and which control are chosen. This pattern is itself the paper’s central point: pair-level bootstrap arbitrarily assigns significance to arbitrary control-metric combinations, while cluster-robust bootstrap reveals that none of the contrasts are individually identifiable.

Table 10: DiD estimates under pair-level (literature convention) vs. two-way cluster bootstrap, on pooled Pearson r . DiD > 0 would indicate pretrained-ESM-2 gains more on the contaminated benchmark relative to clean holdout than control does.

Control	DiD	Pair CI	Pair p	Two-way CI
DeepDTA	+0.217	[+0.137, +0.296]	0.000	[−0.480, +0.574]
Random-proj.	+0.078	[−0.002, +0.160]	0.028	[−0.376, +0.398]
Random-init transf.	+0.194	[+0.111, +0.276]	0.000	[−0.278, +0.629]

The pair-level reading would be: "ESM-2 exhibits a contamination signal against DeepDTA and against the random-init transformer, but not against the random projection probe." The cluster-robust reading is simpler: none of the DiDs is distinguishable from zero at this sample size.

5 Discussion

5.1 Why does a randomly-initialized transformer win?

Three plausible mechanisms. They are not mutually exclusive.

(1) The pretrained probe is actively worse on non-kinase targets. Pretrained ESM-2 representations were trained on UniRef50, which is protein-family-imbalanced. ESM-2 embeddings of non-kinase targets may land in regions of representation space that the MLP head, trained predominantly on kinase data, handles poorly. A randomly-initialised transformer has no such family-specific bias; its embeddings are target-identifier-like without encoding learned family structure that the MLP was not trained to use.

(2) Pretrained embeddings encourage family-specific overfitting. Pretrained ESM-2 embeddings cluster proteins by evolutionary similarity—kinases land near kinases. The MLP head, trained almost entirely on kinase data, learns patterns that exploit this clustering. Those patterns fail on novel-chemistry pairs where the binding is driven by compound structure rather than by the already-seen kinase. A randomly-initialised transformer produces embeddings with no such learned clustering; each target is an arbitrary point in representation space. The MLP has fewer hooks on which to overfit the kinase structure, and correspondingly weaker benchmark performance—but also correspondingly weaker failure on targets outside the training family.

(3) Apparent advantage is small-sample noise within significance. The pairwise $\Delta = +0.138$ has 95% CI with lower bound $+0.004$. This is barely significant; at 20 qualifying targets instead of 12 the significance might dissolve. The finding is real at the sample size we have, but should not be over-read.

None of these mechanisms individually establishes that pretraining is counter-productive for DTI generalisation. Collectively they establish that the apparent *advantage* of pretraining under pooled r on Davis/KIBA is artifactual. A defensible stronger reading requires more datasets, more pretrained checkpoints, and a controlled ablation of pretraining corpus composition.

5.2 What does a DTI benchmark actually measure?

Under cluster-robust inference and macro-averaged metrics, the central finding of this paper is negative: *none* of the four baselines tested generalises meaningfully to novel chemistry on a realistic post-2021 ChEMBL holdout. The best non-trivially positive number is the random-init transformer at macro-avg $r = +0.103$, with 95% CI containing zero.

Published Pearson r values above 0.85 on Davis and KIBA measure predictive ability on a specific, concentrated distribution of kinases and compounds. They do not measure the ability to predict binding affinity for chemistry outside that distribution. This is not a new observation [Wallach and Heifets, 2018, Bender et al., 2022], but the Q92918 case shows that even the pooled holdout numbers—which are supposed to quantify exactly this novel-chemistry behaviour—are substantially driven by single-target effects rather than broad generalisation.

5.3 Actionable recommendations for DTI evaluation

We recommend three concrete reporting changes.

1. **Macro-averaged per-target r as primary metric**, with $n_{\min} \geq 10$ pairs per target. Pooled r should be reported only as a secondary metric, clearly flagged as sensitive to pair-count imbalance.
2. **Per-target pair-count distribution reported in every paper.** A single line—"top-5 targets account for $X\%$ of test pairs"—would make the concentration visible to reviewers and readers.
3. **Two-way (protein $\times$ compound) cluster bootstrap CIs**, or a principled alternative (mixed-effects, cluster-robust sandwich estimator). Pair-level CIs should not be reported as the primary inference procedure on DTI benchmarks.

These changes are additive to existing practice; they do not require re-running any baselines or redesigning any benchmarks.

6 Limitations

Small number of qualifying targets and bootstrap coverage. At $n_{\min} = 10$ the holdout yields only 12 qualifying targets. $K = 12$ is well inside the “danger zone” ($K < 30$) identified by Cameron and Miller [2015] for target-level bootstrap inference. The raw-percentile bootstrap on Pearson r without Fisher- z transformation is known to under-cover at small K [MacKinnon et al., 2023]; under a realistic 85–88% effective coverage, the reported CIs are fragile: the Fisher- z CI $[-0.012, +0.283]$ already includes zero, and the BCa lower bound ($+0.006$) is numerically near-zero. Fisher- z transformation with BCa or studentised bootstrap- t intervals [MacKinnon et al., 2023] would provide more reliable coverage; we report the percentile bootstrap as the primary procedure and note this as a limitation. The diagnostic value of the rank reversal (8 of 12 targets, distributed across protein families) does not depend on formal significance, but formal superiority claims should be treated as indicative rather than conclusive.

External replication and multiplicity. The concentration artifact and random-init-wins finding replicate on TDC DTI-DG (BindingDB_Patent, 190 qualifying targets, Section 4.7). The full rank ordering does not replicate exactly across both datasets. Table 4 and Table 7 scan multiple contrasts and cutoffs without multiplicity correction; a strict Bonferroni adjustment would widen the effective threshold. The diagnostic rank reversal is robust to cutoff choice and leave-one-out perturbation, but the formal pairwise superiority claim rests on a single contrast that barely clears a standard threshold. A companion analysis on other temporal holdouts (BindingDB, TDC DTI-DG, or ChEMBL-based splits used by newer foundation models) would test whether the concentration pattern and rank reversal replicate across data sources; we expect they would, since the underlying cause—long-tailed target distribution in post-cutoff bioactivity data—is not specific to our construction.

Single pretrained model. We evaluate one pretrained configuration: ESM-2 (t33_650M_UR50D) with ChemBERTa (zinc-base-v1). Stronger claims about pretraining broadly would require additional foundation models (MoLFormer, ChemBERTa-3, ESM-3).

Multi-seed random initialisation. We ran the random-init transformer with three seeds {7, 42, 99}. Per-seed macro-avg r values are $+0.103$, $+0.172$, and $+0.078$; multi-seed mean is $+0.138$. All three seeds outperform the pretrained ESM-2 probe (-0.035), and the pooled-prediction macro-avg ($+0.138$) exceeds the single-seed primary result. The variance across seeds (0.078 to 0.172) is itself evidence that the rank advantage is modest rather than structural. Multi-seed predictions are in `results/random_transformer_multiseed_predictions.parquet`.

Cluster bootstrap under-covers at strong cluster structure. At 76% empirical coverage, two-way cluster bootstrap is a substantial improvement over pair-level (17%) but does not hit nominal 95%. With only 12 qualifying targets in the primary macro-avg analysis, we are firmly in the few-cluster regime where standard cluster bootstrap is known to under-reject [MacKinnon and Webb, 2017]. We additionally compute Rademacher wild cluster bootstrap at the target level [MacKinnon and Webb, 2017]: wild CIs for the random-init transformer are $[-0.079, +0.287]$ versus standard target-bootstrap $[-0.102, +0.281]$, slightly narrower because Rademacher perturbation preserves variance structure rather than resampling outlier targets repeatedly. Directional conclusions are unchanged. Mixed-effects models with random target intercepts remain a natural improvement; we report two-way cluster as the minimum-defensible primary procedure. The barely-significant lower CI bound for the pairwise contrast should be interpreted as a diagnostic finding, not a stable superiority claim.

Davis/KIBA pK_d transformation. We convert Davis raw K_d to p K_d before z-scoring to align with KIBA and pChEMBL conventions. Published DeepDTA results using raw K_d may differ numerically from our reproduction; we believe our conversion is the correct choice and have verified that reported trends are consistent with the original DeepDTA [Öztürk et al., 2018].

7 Conclusion

Three evaluation artifacts compound on standard DTI benchmarks: one kinase dominates the test set, pair-level CIs undercover 5×, and pooled metrics hide both. Under proper inference, no model reliably generalises to novel chemistry, and a randomly-initialised transformer outperforms its pretrained counterpart under per-target-equal-weight metrics. The recommendation is simple: macro-averaged per-target Pearson r with cluster-robust CIs, with pooled r as a flagged secondary metric. Whether published DTI rankings survive these corrections is an empirical question the field should be able to answer.

Reproducibility

Code, data splits, and analysis scripts are at <https://github.com/swmeyer1979/dti-contamination>. The full pipeline re-runs from a single shell script. ChEMBL27 and ChEMBL34 source data are publicly available under CC-BY-SA 3.0.

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Competing interests

The author declares no competing financial or non-financial interests.

Data and code availability

All code, processed data splits, and pre-computed model predictions are available at <https://github.com/swmeyer1979/dti-contamination> under an MIT licence. Raw activity data are derived from ChEMBL34 and ChEMBL27, both distributed under CC BY-SA 3.0; derived split files in this repository are correspondingly distributed under CC BY-SA 4.0. Model checkpoints are not archived due to size but are fully reproducible via the provided pipeline script.

References

- A. Bender, N. Schneider, M. H. S. Segler, W. P. Walters, O. Engkvist, and T. Rodrigues. Evaluation guidelines for machine learning tools in the chemical sciences. *Nature Reviews Chemistry*, 6(6):428–442, 2022. doi: 10.1038/s41570-022-00391-9.
- X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, et al. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems (MLSys)*, volume 3, pages 747–769, 2021.

- A. Bushuiev, R. Bushuiev, J. Sedlar, T. Pluskal, J. Damborsky, S. Mazurenko, and J. Sivic. Revealing data leakage in protein interaction benchmarks. *arXiv preprint arXiv:2404.10457*, 2024.
- A. C. Cameron and D. L. Miller. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015.
- A. C. Cameron, J. B. Gelbach, and D. L. Miller. Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29(2):238–249, 2011.
- L. Chen, A. Cruz, S. Ramsey, et al. Hidden bias in the DUD-E dataset leads to misleading performance of deep learning in structure-based virtual screening. *PLOS ONE*, 14(8):e0220113, 2019.
- S. Chithrananda, G. Grand, and B. Ramsundar. ChemBERTa: large-scale self-supervised pre-training for molecular property prediction. *arXiv preprint arXiv:2010.09885*, 2020.
- M. I. Davis, J. P. Hunt, S. Herrgard, et al. Comprehensive analysis of kinase inhibitor selectivity. *Nature Biotechnology*, 29(11):1046–1051, 2011.
- B. Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.
- K. Huang, C. Xiao, L. M. Glass, and J. Sun. MolTrans: molecular interaction transformer for drug–target interaction prediction. *Bioinformatics*, 37(6):830–836, 2021.
- K.-Y. Liang and S. L. Zeger. Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1):13–22, 1986.
- Z. Lin, H. Akin, R. Rao, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.
- J. G. MacKinnon and M. D. Webb. Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254, 2017.
- J. G. MacKinnon, M. Ø. Nielsen, and M. D. Webb. Fast and reliable jackknife and bootstrap methods for cluster-robust inference. *Journal of Applied Econometrics*, 38(5):671–694, 2023. doi: 10.1002/jae.2969.
- T. Nguyen, H. Le, T. P. Quinn, et al. GraphDTA: predicting drug–target binding affinity with graph neural networks. *Bioinformatics*, 37(8):1140–1147, 2021.
- H. Öztürk, A. Özgür, and E. Ozkirimli. DeepDTA: deep drug–target binding affinity prediction. *Bioinformatics*, 34(17):i821–i829, 2018.
- T. Pahikkala, A. Airola, S. Pietilä, S. Shakyawar, A. Szwejda, J. Tang, and T. Aittokallio. Toward more realistic drug–target interaction predictions. *Briefings in Bioinformatics*, 16(2):325–337, 2015. doi: 10.1093/bib/bbu010.
- J. Ross, B. Belgodere, V. Chenthamarakshan, et al. Large-scale chemical language representations capture molecular structure and properties. *Nature Machine Intelligence*, 4(12):1256–1264, 2022.
- A. Søgaard, A. Johannsen, B. Plank, D. Hovy, and H. M. Alonso. What’s in a p-value in NLP? In *Proceedings of the 18th Conference on Computational Natural Language Learning (CoNLL)*, pages 1–10, 2014.

- J. Tang, A. Szwajda, S. Shakyawar, et al. Making sense of large-scale kinase inhibitor bioactivity data sets: a comparative and integrative analysis. *Journal of Chemical Information and Modeling*, 54(3):735–743, 2014.
- K. Vishniakov, K. Viswanathan, A. Medvedev, P. Kanithi, M. A. F. Pimentel, R. Rajan, and S. Khan. Genomic foundationless models: Pretraining does not promise performance. *bioRxiv*, 2024. doi: 10.1101/2024.12.18.628606.
- E. M. Voorhees. Overview of the TREC 2004 robust retrieval track. In *Proceedings of the Thirteenth Text REtrieval Conference (TREC 2004)*. NIST Special Publication 500-261, 2004.
- I. Wallach and A. Heifets. Most ligand-based classification benchmarks reward memorization rather than generalization. *Journal of Chemical Information and Modeling*, 58(5):916–932, 2018.
- Z. Xu, R. Gupta, W. Cheng, A. Shen, J. Shen, A. Talwalkar, and M. Khodak. Specialized foundation models struggle to beat supervised baselines. In *Proceedings of the Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
- B. Zdrazil, E. Felix, F. Smart, et al. The ChEMBL database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1): D1180–D1192, 2024.